Divyansh Shukla

+91 7827910144 | github.com/divyanshuklai | linkedin.com/in/divyansh-shukla-ai | x.com/divshkl

Summary

Machine Learning Engineer (Systems) with a strong foothold in data pipeline management, model training/fine-tuning, and ML on the edge, with projects spanning flow-matching/diffusion, decoder-only transformers, vision-language model (VLM) experimentation, and robot learning.

Education

Vellore Institute of Technology
Bachelor of Technology in Computer Science and Engineering (Specialization: AI & ML) — *CGPA: 9.03*

DAV Public School Sahibabad
XII Standard — *91%*

DAV Public School Sahibabad
X Standard — *95.6%*

Skills

Languages: Python, C++, CUDA, HTML
Libraries: PyTorch (w/ Distributed), PyTorch Lightning, Hugging Face Transformers, TRL, vLLM, pandas, OpenCV
RL/Sim: Gymnasium, MuJoCo
ML/DL: diffusion models, flow matching, transformer architectures, reinforcement learning, vision-language model experimentation
Tools/Workflow: Hydra, Jupyter notebooks, Modal (remote compute)

Projects

VLM Project | github.com/divyanshuklai/vlm-project

- Built an end-to-end VLM pipeline (data prep → training → eval) that simplified experiment iteration and reduced time-to-result.
- Implemented alignment + preference optimization methods including RLHF (reward modeling + optimization), DPO, and RLVR (GRPO-style RL fine-tuning).
- Enhanced evaluation rigor with benchmark-driven comparisons (competitive with similarly sized models) and standardized, reproducible runs on Modal.

Zero-shot ICE RMA (Teacher–Student RL) | github.com/divyanshuklai/zero-shot-ice-rma

- Replicated the original RMA (teacher–student) approach and extended it into a reproducible training + evaluation workflow.
- Increased robustness and generalization via built-in domain randomization and evaluation across a multitude of environments for training and transfer testing.
- Enabled easier sim-to-sim transfer and laid groundwork for sim-to-real deployment tooling (repeatable configs, eval harness, and deployment-oriented utilities).

Fashion Search | github.com/divyanshuklai/fashion_search

- Built a FastAPI conversational chat app for natural-language fashion product discovery across 10 Indian e-commerce stores (Myntra, Ajio, TataCliq, and others).
- Implemented a naive search pipeline (Tavily web search → Gemini 2.5 Flash parsing → BeautifulSoup scraping) to turn search snippets into structured product data.
- Engineered a priority-based image-extraction strategy (JSON-LD, meta tags, keyword-matched img tags, OG fallback) to reliably surface product images and prices in a chat UI; extending to an agentic search variant with planning, memory, and tool use for comparison against the naive baseline.

Relevant Knowledge

Applied Accelerated Artificial Intelligence (NPTEL25CS98)
Deep Learning with Computer Vision (NPTEL25CS93)
Deep Learning Specialization (Coursera)